

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 2

What Numbers Actually Are

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. Someone says: “Imaginary numbers aren’t real, they’re just a trick.” In what precise sense are imaginary numbers exactly as “real” as real numbers? In what sense is the word “imaginary” misleading?

2. Why can’t you put the complex numbers in order the way you can put the reals in order? Is this a deficiency or a feature?

3. You encounter a system with two elements and an operation that combines them. Under what conditions would a mathematician say this system is “structurally identical” to something you already know?

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4. What is the structural reason that complex numbers appear in quantum mechanics and electrical engineering but not in accounting?

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5. A colleague says: “Infinity is infinity, talk of ‘bigger’ and ‘smaller’ infinities is philosophy.” Using the pairing definition of size, state what it would even *mean* for one infinite set to be strictly bigger than another, and name two infinite sets that are provably different sizes. What single maneuver does the proof of that difference rest on?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
